

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12410568
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Owens; Matt

Shallow foundation bollard structures

Abstract

A bollard structure includes a base and a bollard. The base includes an upper wall, a lower wall, a first side wall extending between the upper and lower walls, a second side wall extending between the upper and lower walls and spaced apart from the first side wall, and a mid-wall extending between the upper and lower walls and between the first and second sidewalls. The bollard penetrates the upper wall and extends to the lower wall between the two sidewalls, such that the mid-wall extends from the bollard.

Inventors:	Owens; Matt (Palmdale, CA)
Applicant:	Delta Scientific Corporation (Palmdale, CA)
Family ID:	85288646
Assignee:	DELTA SCIENTIFIC CORPORATION (Palmdale, CA)
Appl. No.:	17/709120
Filed:	March 30, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230062533 A1	Mar. 02, 2023

Related U.S. Application Data

us-provisional-application US 63238738 20210830

Publication Classification

Int. Cl.: E01F15/00 (20060101)

U.S. Cl.:

CPC **E01F15/003** (20130101);

Field of Classification Search

CPC: E01F (9/685); E01F (13/12); E01F (15/003)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1449518	12/1922	Lawson	N/A	N/A
3934540	12/1975	Bruner et al.	N/A	N/A
4490068	12/1983	Dickinson	N/A	N/A
5228237	12/1992	Nasatka	N/A	N/A
7104721	12/2005	Rogers et al.	N/A	N/A
7118304	12/2005	Turpin et al.	N/A	N/A
7699558	12/2009	Adler et al.	N/A	N/A
7775738	12/2009	Darcy	404/6	E01F 13/12
7850391	12/2009	Omar	404/6	E01F 13/12
8215865	12/2011	Adler et al.	N/A	N/A
10988903	12/2020	Neusch	N/A	E01F 15/003
11174606	12/2020	Lamore	N/A	E01F 13/026
2009/0028638	12/2008	Crawford	404/6	E01F 13/12

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
200071449	12/2000	AU	N/A
2485278	12/2011	GB	E01F 13/12
2600120	12/2021	GB	E01F 13/048
2002-115324	12/2001	JP	N/A

OTHER PUBLICATIONS

DECISION Granting Institution of Inter Partes Review issued by the USPTO Patent Trial and Appeal Board, dated Feb. 5, 2021, 59 pages. cited by applicant

DECISION Denying Institution of Inter Partes Review issued by the USPTO Patent Trial and Appeal Board, dated Nov. 21, 2019, 34 pages. cited by applicant

Primary Examiner: Hartmann; Gary S

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This patent application claims the benefits of U.S. Provisional Patent Application Ser. No. 63/238,738, filed on Aug. 30, 2021, and

entitled "SHALLOW FOUNDATION BOLLARD STRUCTURES" the entire content of which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Vehicle barriers such as bollards are well known in the art. Shallow mounted bollards are used in situations where the bollards need to be mounted above utilities. Most shallow mounted bollards have a base which extends horizontally formed by multiple crossing tubular members. Often multiple bollards are mounted on such bases. As such the configurations and spacing between bollards is set. A shallow mounted bollard structure is desired that can be mounted independently of other bollard structures in close proximity.

SUMMARY

(2) In an example embodiment, a bollard structure includes a base and a bollard. The base includes an upper wall, a lower wall, a first side wall extending between the upper and lower walls, a second side wall extending between the upper and lower walls and spaced apart from the first side wall, and a mid-wall extending between the upper and lower walls and between the first and second sidewalls. The bollard penetrates the upper wall and extends to the lower wall between the two sidewalls, such that the mid-wall extends from the bollard. In another example embodiment, the bollard structure further includes at least one upper reinforcing plate above the upper wall penetrated by the bollard. In a further example embodiment, the bollard structure also includes at least one lower reinforcing plate penetrated by the bollard. In one example embodiment, the mid-wall is a web of an I-beam, such that the I-beam extends from the upper wall to the lower wall. In another example embodiment, the mid-wall is a first mid-wall, and the structure further includes a second mid-wall extending from the bollard opposite from the first mid-wall. In yet another example embodiment, the second mid-wall is welded to the bollard. In a further example embodiment, the second mid-wall is a web of a second I-beam, such that the second I-beam extends from the upper wall to the lower wall. In yet a further example embodiment, the upper wall is a web of an upper channel. The upper channel includes a first wall and second wall extending transversely from the web, such that the first side wall is connected to the upper channel first wall and the second side wall is connected to the upper channel second wall. In one example embodiment, the lower wall is a web of a lower channel. The lower channel includes a first wall and second wall extending transversely from the lower channel web, such that the first side wall is connected to the lower channel first wall and the second side wall is connected to the lower channel second wall. In another example embodiment, the bollard is tubular, and the bollard structure further includes a stiffener extending within the bollard. In yet another example embodiment, the stiffener extends from a base of the bollard along at least 80% of a length of the bollard. In a further example embodiment, the stiffener is a box beam. In yet a further example embodiment, the stiffener is a plate. In one example embodiment, the stiffener is connected to opposite inner surfaces of the bollard. In another example embodiment, the first side wall includes first openings formed therethrough. In yet another example embodiment, the second side wall includes second openings formed therethrough. In a further example embodiment, the first and/or second openings are sized to receive a fork of a fork lift allowing for lifting and movement of the structure by the fork lift. In yet a further example embodiment, the first and/or second openings are sized to allow for a person to weld and inspect an interior of the bollard structure. In an example embodiment, the first and second openings are configured for receiving rebar bundles therethrough. In another example embodiment, the mid-wall includes openings axially aligned with the first and second openings for receiving rebar bundles therethrough. In yet another example embodiment, the bollard structure further includes a rebar cage surrounding the base. In a further example embodiment, the rebar cage includes a lower rebar grid and an upper rebar grid, such that the base is sandwiched between the lower and upper rebar grids. The rebar cage also includes a plurality of spaced apart truncated cone shaped rebar members extending from the upper wall to the lower rebar grid, and a

plurality of hook shaped rebar members. In an example embodiment, each hook shaped rebar member of the plurality of hook shaped rebar members includes a hook portion hooked on the upper rebar grid and extending to the lower rebar grid. In yet a further example embodiment, the plurality of hook shaped rebar members are connected to the first and second rebar grids and the plurality of truncated cone shaped rebar members are connected to the lower grid.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective assembly view of an example embodiment bollard structure.
- (2) FIG. 2 is a perspective view of a partial assembled view of the bollard structure shown in FIG. 1.
- (3) FIG. 3 is a side view of the bollard structure shown in FIG. 1.
- (4) FIG. 4 is a perspective view of an example embodiment bollard structure with a rebar structure shown in assembly view.
- (5) FIG. 5 is a perspective view of the example embodiment bollard structure shown in FIG. 4 with the rebar structure shown in FIG. 4.

DESCRIPTION

- (6) An example shallow foundation bollard structure **10** has a base **12** from which extends a bollard **14** as for example shown in FIGS. 1, 2 and 3. In an example embodiment, the bollard is a hollow tubular bollard. The tubular bollard may have a cross-sectional shape that is circular, oval, square, rectangular, or any other geometric shape. In the shown example embodiment in FIG. 1, the tubular bollard has a circular cross-sectional shape, thus defining a circular tubular member. The base is a box beam formed by an upper channel **16**, a lower channel **18**, a first I-beam **20**, a second I-beam **22** spaced apart from the first I-beam to receive a bollard **14** therebetween, and a first side wall **24** opposite a second side wall **26**.
- (7) The first I-beam **20** has a first upper cap **20A**, a lower cap **20B** and a web **20C** extending between the first I-beam upper and lower caps. Similarly, the second I-beam **22** has an upper cap **22A**, a lower cap **22B** and a web **22C** extending between the second I-beam upper and lower caps. In other example embodiments, a vertical plate, or back to back channels defining an I-beam, or a box beam having an upper wall, a lower wall and two side walls extending between the upper and lower walls may be used instead of each I-beam. For illustrative purposes embodiments herein are described with using I-beams.
- (8) The upper channel **16** includes a web **16A** and first and second walls **16B**, **16C**, extending transversely from the web defining the upper channel. The lower channel **18** includes a web **18A** and first and second walls **18B**, **18C**, extending transversely from the lower channel web defining the lower channel. First and second upper reinforcing plates **28A** and **28B**, respectively, and first and second lower reinforcing plates **29a** and **29B**, respectively, may be attached above the upper channel and below the lower channel, respectively to reinforce the attachment of the bollard to the base.
- (9) The upper and lower channels define the upper and lower ends or walls of the box beam (i.e., the base). Each side wall, which in the shown example embodiment is a plate, is attached to both the upper channel and the lower channel such that the first side wall **24** forms a first side wall of the box beam while the second side wall **26** forms a second side wall of the box beam opposite the first side wall. The first and second I-beams are also welded to the upper and lower channels. In an example embodiment, an end **30** of the first I-beam facing the second I-beam, and an end **32** of the second I-beam facing the first I-beam, have curvatures complementary to the outer curvature of the bollard. Specifically, the upper and lower caps of each I-beam have a curvature complementary the outer surface curvature of the bollard. Openings **36A**, **36B** are formed through the upper and lower

channel webs, respectively, to receive the bollard. Openings **38A**, **38B** are also formed through the upper reinforcing plates **28A**, **28B**, respectively, and openings **39a**, **39b** (not shown), are formed through the lower reinforcing plates **29B**, **29A**, respectively, to receive the bollard.

(10) The bollard **14** penetrates the openings **38A**, **38B** of the upper reinforcing plates, the opening **36A** of the upper channel **16**, the spacing between the two I-beams, the opening **36B** of the lower channel **18** and the openings **39B**, **39A** of the lower reinforcing plates. In an example embodiment, the bollard is welded to the upper reinforcing plates, the opening of the upper channel, both I-beams, the opening of the lower channel and the openings of the lower reinforcing plates.

(11) Axially aligned openings **40**, **42**, **44** are formed through the first side wall **24**, the I-beams **20**, **22** and the second side wall **26**, respectively, to accommodate rebar or bundles of rebar **70**. In an example embodiment, the openings are also sized to receive the arms (i.e., the forks) of a fork lift, allowing for manipulation and movement of the bollard structure by a fork lift. In addition, the opening may have sufficient size to allow for welding within the base, as for example, for welding the inner surfaces of the side walls to the upper and lower channel first and second walls, and if needed welding the bollard to the I-beams as well as welding of the I-beams to the upper and lower channels. The opening **40**, **42**, **44** can have various shapes as for example circular as shown in FIG. **1**, oval as shown in dashed is FIG. **3** and in solid in FIG. **4**, or another shape that would allow for welding and/or accommodation of the arms of a fork lift.

(12) As shown in FIG. **2**, to form the bollard structure **10**, the bollard **14** is placed over the lower channel **18**, such that a lower end of the bollard penetrates the opening **36B** of the lower channel and extends the thickness of the lower reinforcing plates **29B**, **29A**. The first I-beam **20** is then placed on the lower channel web inner surface along a central longitudinal axis of the lower channel such that its upper and lower curved cap ends **30** and its web **20C** abut the outer surface of the bollard. Similarly, the second I-beam **22** is placed on the lower channel web inner surface along the central longitudinal axis of the lower channel such that upper and lower cap curved ends **32** and its web **22C** abut the outer surface of the bollard opposite the first I-beam. In an example embodiment, the I-beams are welded to the lower channel member and to the bollard.

(13) The upper channel **16** is placed over the bollard, such that the bollard penetrates the upper channel opening **36A**. In an example embodiment, the upper channel is welded to the upper caps of the I-beams. The lower reinforcing plates **29B**, **29A**, are placed over the outer surface of the web **18A** of the lower channel such that they are penetrated by the bollard lower end and are longitudinally aligned with the lower channel. Similarly, the upper reinforcing plates **28B**, **28A**, are placed over the outer surface of the web **16A** of the upper channel such that they are penetrated by the bollard end and are longitudinally aligned with the upper channel. In an example embodiment, the lower reinforcing plates are welded to each other, the bollard and to the lower channel, the upper reinforcing plates are welded to each other, the bollard and the upper channel.

(14) The first and second sidewalls **24**, **26** are then welded to the vertically extending walls of the upper and lower channels. For example, the first sidewall is welded to the first walls **16B**, **18B** of the upper and lower channels, respectively, and the second sidewall is welded to the second walls **16C**, **18C**, of the upper and lower channels, respectively. The first and second sidewall may be welded in the outer surfaces of their corresponding channel vertical walls as shown in FIGS. **3** and **4** or in another example embodiment, may be welded to inner surface of their corresponding channel vertical walls.

(15) In an example embodiment, a stiffener **50** may be inserted in the bollard that spans a majority of the bollard length and preferably at least 80% of the bollard length extending from the base of the bollard, as for example shown in FIGS. **1** and **3**. In an example embodiment, the stiffener is a box beam, as for example shown in FIGS. **1** and **3**. In an example embodiment, the box beam stiffener is rectangular in cross-section and is welded to opposite interior surfaces of the bollard with its longer sides **52** parallel or generally parallel to the direction of expected impact **54**, as for example shown in FIG. **3**. In another example embodiment, the stiffener **50** may be a plate, that

extends to opposite interior surfaces of the bollard. With such embodiment, the plate is welded to opposite interior surfaces of the bollard, such that a plane of the plate is parallel or generally parallel to the direction of expected impact. In other example embodiments, the stiffener may have other geometric shapes. A box stiffener is expected provide more bending resistance to the bollard than a plate stiffener.

(16) In an example embodiment as shown in FIG. 4, during installation of the bollard structure, a first or lower rebar grid **56** is placed on the base of a shallow excavated pit. The shallow foundation bollard structure **10** is placed above the first rebar grid. In an example embodiment, truncated cone shaped rebar members **58** are placed over the base and rest on the base. Longitudinally extending rebar members **59** may be connected, as for example by welding, to the truncated cone shaped rebar members forming a truncated cone shaped rebar member grid. In an example embodiment, the truncated cone shaped rebar members are welded, tied or otherwise connected to the first rebar grid. A second or upper rebar grid **60** is placed above the bollard structure and is penetrated by the bollard **14** of the bollard structure. The second rebar grid may be connected to the truncated cone shaped rebar members. Hook shaped rebar members **62** defining a hook **64** are hung with their hooks on the second rebar grid and extend to the first rebar grid, as for example shown in FIG. 5. The hook shaped rebar members are connected to the first and second rebar grid. In an example embodiment, they may be tied or welded to the first and second rebar grids. Since the second rebar grid will be very close to the upper surface of the excavated pit after it is filled with concrete, there isn't much room for rebar overlap. Each hook shaped rebar member hook helps prevent the concrete from shearing or breaking, at or proximate such surface, in tension at the intersection of the hook and the second rebar grid. The first and second rebar grids, along with the truncated cone shaped rebar members and the hook shaped rebar members form rebar cage **68** surrounding the base **12**. Rebar or rebar bundles **70** are placed and extend transversely through the base **12** of the bollard structure through the openings **40, 42, 44**. In example embodiments, the rebar bundles are connected to the rebar cage **68**, as for example by being welded or tied to the rebar cage. Concrete (or other retaining media) is poured over the base to fill the excavation site and encapsulate the rebar cage. In an example embodiment, concrete is also poured into the bollard. A cap **72** may be used to cap the upper end of the bollard after it is filled with concrete.

(17) In example embodiments, multiple shallow foundation bollard structures may be installed spaced apart from each other with rebar or rebar bundles **70** penetrating all such bollard structures. In an example embodiment, a single cage may be formed surrounding all bollard structures. Example embodiment shallow foundation bollards may be installed at excavated pits less than three feet in depth from grade. In other example embodiments, the excavated pits may be up to two feet in depth from grade. In a further example embodiment, the excavated pits are about two feet in depth from grade.

(18) In an example embodiment, upon impact, the base with the bollard will rotate such that the base impacts the undercarriage of the impacting vehicle, further aiding the stopping of the vehicle. The example embodiment foundation bollards described herein have been able to stop a 15,000 truck having a velocity of 52 MPH impacting the bollard structure along direction **54** as shown in FIG. 3 and meet the 2020 ASTM M50 Standard P2 Dynamic and P1 Static requirements as well as the 2020 ASTM M30 P1 requirements. Similar results are expected if the impact direction is opposite of the direction **54** shown in FIG. 3.

(19) While this invention has been described in detail with particular references to exemplary embodiments thereof, the exemplary embodiments described herein are not intended to be exhaustive or to limit the scope of the invention to the exact forms disclosed. Persons skilled in the art and technology to which this invention pertains will appreciate that alterations and changes in the described structures and methods of assembly and operation can be practiced without meaningfully departing from the principles, spirit, and scope of this invention, as set forth in the following claims. For example, the upper and lower walls of the base may be formed by plates

instead of channel members and the two side walls may be formed by channel members instead of plates. In other example embodiments, the upper, lower and side walls may be formed from plates that are welded or otherwise interconnected together. In one example embodiment, corner members may be used to connect or weld such adjacent plates.

(20) Although relative terms such as “outer,” “inner,” “upper,” “lower,” “below,” “above,” and similar terms have been used herein to describe a spatial relationship of one element to another, it is understood that these terms are intended to encompass different orientations of the various elements and components of the invention in addition to the orientation depicted in the figures. Additionally, as used herein, the term “generally,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent deviations in measured or calculated values that would be recognized by those of ordinary skill in the art. Furthermore, as used herein, when a component is referred to as being “on” another component, it can be directly on the other component or components may also be present therebetween. Moreover, when a component is referred to as being “coupled” to another component, it can be directly attached to the other component or intervening components may be present therebetween.

Claims

1. A bollard structure comprising: a base comprising, an upper wall, a lower wall, a first side wall extending between the upper and lower walls, a second side wall extending between the upper and lower walls and spaced apart from the first side wall, and a mid-wall extending between the upper and lower walls and between the first and second sidewalls; and a bollard penetrating the upper wall and extending to the lower wall between the two sidewalls, wherein said mid-wall extends from said bollard, wherein the mid-wall is a web of an I-beam, wherein the I-beam extends from the upper wall to the lower wall.
2. The structure of claim 1, further comprising at least one upper reinforcing plate above the upper wall penetrated by said bollard.
3. The structure of claim 1, further comprising at least one lower reinforcing plate penetrated by said bollard.
4. The structure of claim 1, wherein said mid-wall is a first mid-wall, and the structure further comprises a second mid-wall extending from the bollard opposite from said first mid-wall.
5. The structure of claim 4, wherein the second mid-wall is welded to the bollard.
6. The structure of claim 4, wherein the second mid-wall is a web of a second I-beam, wherein the second I-beam extends from the upper wall to the lower wall.
7. The structure of claim 1, wherein the upper wall is a web of an upper channel, said upper channel comprising a first wall and second wall extending transversely from said web, wherein the first side wall is connected to the upper channel first wall and the second side wall is connected to the upper channel second wall.
8. The structure of claim 4, wherein the lower wall is a web of a lower channel, said lower channel comprising a first wall and second wall extending transversely from said lower channel web, wherein the first side wall is connected to the lower channel first wall and the second side wall is connected to the lower channel second wall.
9. The structure of claim 1, wherein the bollard is tubular, the structure further comprising a stiffener extending within the bollard.
10. The structure of claim 9, wherein the stiffener extends from a base of the bollard along at least 80% of a length of the bollard.
11. The structure of claim 9, wherein the stiffener is a box beam.
12. The structure of claim 9, wherein the stiffener is a plate.
13. The structure of claim 9, wherein the stiffener is connected to opposite inner surfaces of the bollard.

14. The structure of claim 1, wherein the first side wall comprises first openings formed therethrough.
 15. The structure of claim 14, wherein the second side wall comprises second openings formed therethrough.
 16. The structure of claim 15, wherein the first and/or second openings are sized to receive a fork of a fork lift for allowing for lifting and movement of said structure by a fork lift.
 17. The structure of claim 15, wherein the first and/or second openings are sized to allow for welding and inspection of an interior of said structure.
 18. The structure of claim 17, wherein the first and second openings are configured for receiving rebar bundles therethrough.
 19. The structure of claim 18, wherein the mid-wall comprises openings axially aligned with the first and second openings for receiving rebar bundles therethrough.
 20. The structure of claim 1, further comprising a rebar cage surrounding said base.
 21. The structure of claim 20, wherein the rebar cage comprises: a lower rebar grid; an upper rebar grid, wherein the base is sandwiched between the lower and upper rebar grids; a plurality of spaced apart truncated cone shaped rebar members extending from the upper wall to the lower rebar grid; and a plurality of hook shaped rebar members, wherein each hook shaped rebar member of said plurality of hook shaped rebar members comprises a hook portion hooked on the upper rebar grid and extending to the lower rebar grid.
 22. The structure of claim 21, wherein the plurality of hook shaped rebar members are connected to upper and lower rebar grids and wherein the plurality of truncated cone shaped rebar members are connected to the lower grid.
 23. A bollard structure comprising: a base comprising, an upper wall, a lower wall, a first side wall extending between the upper and lower walls, a second side wall extending between the upper and lower walls and spaced apart from the first side wall, and a mid-wall extending between the upper and lower walls and between the first and second sidewalls; and a bollard penetrating the upper wall and extending to the lower wall between the two sidewalls, wherein said mid-wall extends from said bollard, wherein said mid-wall is a first mid-wall, and the structure further comprises a second mid-wall extending from the bollard opposite from said first mid-wall, and wherein the second mid-wall is a web of an I-beam, wherein the I-beam extends from the upper wall to the lower wall.
 24. A bollard structure comprising: a base comprising, an upper wall, a lower wall, a first side wall extending between the upper and lower walls, a second side wall extending between the upper and lower walls and spaced apart from the first side wall, and a mid-wall extending between the upper and lower walls and between the first and second sidewalls; a bollard penetrating the upper wall and extending to the lower wall between the two sidewalls, wherein said mid-wall extends from said bollard; and at least one upper reinforcing plate above the upper wall penetrated by said bollard.
 25. The structure of claim 24, further comprising at least one lower reinforcing plate penetrated by said bollard.
 26. A bollard structure comprising: a base comprising, an upper wall, a lower wall, a first side wall extending between the upper and lower walls, a second side wall extending between the upper and lower walls and spaced apart from the first side wall, and a mid-wall extending between the upper and lower walls and between the first and second sidewalls; a bollard penetrating the upper wall and extending to the lower wall between the two sidewalls, wherein said mid-wall extends from said bollard, and at least one lower reinforcing plate penetrated by said bollard.
-